Solution Requirements

Date	28 June 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:
Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The web application does not require user login. Any user can directly access the home page and prediction page without authentication since it is an open educational tool .	Low

2	Authorization levels	All users have equal access to all pages — Home, Prediction form, and Result page. No admin or restricted roles are required for this project	Low
3	External interfaces	The system connects to the UNDP HDI dataset (HDI.csv from Kaggle) as an external data source. The trained model file HDI.pkl is loaded by Flask at startup to serve predictions	High
4	Transactions processing	When the user submits the prediction form, the Flask backend reads all four input values, converts them to float, creates a Pandas DataFrame, and passes it to the ML model for prediction in real time	High
5	Reporting	The system displays the predicted HDI score (e.g., 0.48) and its category (e.g., Medium HDI) on the result page. The live estimator on the form page also shows a real-time estimate before final submission	High
6	Business Rules	Input values must be within valid ranges: Life Expectancy 50–89, Expected Schooling 1–18, Mean Schooling 1–15, GNI 600–75000. If the predicted score falls outside 0.30–0.94, the system shows an out-of-range message	High
7	Compliance to laws or regulations	The project uses open UNDP data which is publicly available. No personal user data is collected or stored.	Medium

8	Other	The system must correctly classify the predicted HDI score into four categories: Low (0.30–0.40), Medium (0.40–0.70), High (0.70–0.80), Very High (0.80–0.94) using if-elif conditions in Flask	Medium

Step 2: Non-Functional Requirements (NFR)

S.NO	NFR Category	Requirments Description	Target Metric/Acceptance Criteria
1	Performance & Speed	The Flask web application must load pages quickly and return prediction results without noticeable delay	Prediction result displayed in less than 2 seconds after form submission
2	Scalability	The application should handle multiple users accessing the prediction form simultaneously without crashing	Supports at least 50 concurrent users on a local server environment
3	Security & Data Privacy	No user personal data is collected or stored. All inputs are processed in memory only and discarded after prediction	No user data stored in database or log files
4	Reliability & Availability	The Flask application should run stably during demo and evaluation sessions without crashes or errors	99% uptime during project demonstration period
5	Usability & Accessibility	The web interface must be easy to use with clear labels, input hints, range guidance, and a visually clean dark-themed design	Users can complete a prediction in under 1 minute without guidance

6	Other	The code must be well-commented and organized across separate files — notebook for training, app.py for Flask, and HTML templates — for easy maintenance and handover	Each file clearly separated with comments explaining every section